

TOLEDO CGS, OH, USA

WMO: 997295

Lat: 41.694N

Lon: 83.473W

Elev: 184

StdP: 99.13

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.2	-13.4	N/A	N/A	N/A	N/A	N/A	N/A	11.4	-0.9	10.5	-3.2	3.6	260	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	32.5	N/A	30.8	N/A	29.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.2	8.2	-19.1	35.0	4.3	1.7	-22.2	36.2	-24.8	37.2	-27.2	38.1	-30.3	39.4		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-3.1	-2.7	2.7	9.9	16.2	21.5	23.8	22.7	18.8	12.6	5.4	0.5		
	DBStd	10.63	6.16	5.60	5.49	4.82	4.78	3.70	3.24	2.88	4.01	4.44	5.02	4.96		
	HDD10.0	1537	408	356	238	58	4	0	0	0	22	154	296			
	HDD18.3	3283	666	589	486	255	102	12	1	2	42	187	389	553		
	CDD10.0	1810	1	1	11	56	196	344	427	394	263	101	15	2		
	CDD18.3	515	0	0	0	3	35	106	170	138	55	8	0	0		
	CDH23.3	3736	0	0	2	38	249	744	1402	919	335	47	0	0		
	CDH26.7	1125	0	0	0	3	67	238	459	244	103	10	0	0		
Wind	WSAvg	3.9	4.4	4.2	4.1	4.8	4.0	3.8	3.3	3.1	3.1	3.6	4.1	4.1		
Precipitation	PrecAvg	854	48	46	62	82	88	91	87	86	79	60	62	62		
	PrecMax	1185	92	122	145	127	154	226	160	210	159	160	169	101		
	PrecMin	619	14	9	14	32	14	38	30	19	10	11	22	29		
	PrecStd	133	24	27	34	30	42	45	32	39	35	33	32	21		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	14.1	21.8	27.1	31.1	34.0	34.9	32.9	32.8	27.9	20.1	15.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	10.9	10.1	17.5	24.0	28.7	31.7	32.7	31.0	29.8	24.1	17.1	12.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	8.4	6.8	14.4	21.0	26.4	29.7	30.9	29.4	27.0	21.7	15.1	10.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	5.2	4.2	10.9	18.2	23.8	27.5	29.2	27.8	24.9	19.4	13.0	7.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.5	7.3	8.3	8.3	7.7	8.0	7.7	7.7	7.6	6.7	5.5		
		MCDBR	9.2	9.7	11.7	13.0	11.2	10.4	9.9	9.6	10.6	10.3	10.0	8.6		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Clear-Sky Solar Irradiance	taub		0.301	0.335	0.360	0.405	0.435	0.451	0.444	0.435	0.387	0.362	0.324	0.314		
	taud		2.411	2.297	2.318	2.218	2.182	2.184	2.225	2.234	2.346	2.399	2.465	2.402		
	Ebn at Noon		849	863	884	865	844	829	831	827	845	825	815	796		
	Edh at Noon		79	104	115	136	145	145	138	133	111	93	75	74		
All-Sky Solar Radiation	RadAvg		1.53	2.44	3.49	4.63	5.42	6.05	6.07	5.32	4.32	2.85	1.82	1.25		
	RadStd		0.16	0.32	0.32	0.39	0.47	0.41	0.34	0.20	0.35	0.23	0.24	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+93	N/A	

Nomenclature: See separate page